Comments on "The three-dimensional core–redundant case"

Accuracy check, with focus on the proof of Proposition 1 :: August 28, 2026

Summary

The mathematics in the note is sound. I verified every formal argument step by step and recomputed all numerical claims independently. The proof of Proposition 1 (admissibility of finite rectangles) is correct as written: the zero-mass step, the three first-moment identities, the exponential-tilt argument killing negative comparison mass above the core cutoffs, and the final X_3-moment contradiction all go through. The comments below are of three kinds: (i) places in the proof of Proposition 1 where a hypothesis is used silently and one sentence would make the argument airtight; (ii) one substantive restriction missing from Proposition 2 ($\alpha < 1/2$); and (iii) a small gap in the proof of Proposition 4 (positivity of the transfer sets) plus minor items. None of these affects the correctness of the stated results, except that Proposition 2 as stated is false for $\alpha \geq 1/2$ and should carry the restriction explicitly.

1 Proposition 1: verification and suggestions

Steps verified. (a) $0 \in \Lambda_0$, so $\mathrm{E}_0\psi \leq \alpha = \mathrm{E}_0\phi_c$; the ray te_1 lies in Λ_1 for all $t > 0$ and power functions of tests under Gaussian shifts are continuous, so $\mathrm{E}_0 h = 0$. (b) The identity $\mathrm{E}_{td}h = \exp\{-t^2\|d\|^2/2\}\,\mathrm{E}_0[h\,e^{td'X}]$ is correct, and since $g(t) := \mathrm{E}_0[h\,e^{td'X}]$ satisfies $g(0) = 0$ and $g(t) \geq 0$ for $t > 0$, the right derivative gives $d'q \geq 0$. (c) The directions $(1, s, 0)$ and $(s, 1, 0)$ do belong to Λ_1 for every real s (the max restriction never binds since the third coordinate is zero), and they force $q_1 = q_2 = 0$; the directions $(1, 0, r)$, $r \leq 1$, then force $q_3 = 0$. (d) The support inclusions in (10) hold pointwise, not merely up to null sets: on A_c, $h = \psi \geq 0$; on A_c^c, $h = \psi - 1 \leq 0$. (e) The tilt bound $\mathrm{E}_0[e^{tX_j}h] \leq e^{tc_j}M^+ - e^{t(c_j+\varepsilon)}m_{j,\varepsilon}$ is correct and the union over $\varepsilon = 1/n$ delivers $h^- = 0$ a.s. on $\{X_j > c_j\}$ for $j = 1, 2$. (f) The final display is correct: $\mathrm{E}_0[X_3h^+] \leq c_3M$ because $X_3 \leq c_3$ on the support of h^+, while $\mathrm{E}_0[X_3h^-] > c_3M$ strictly when $M > 0$ because $X_3 > c_3$ a.s. on $\{h^- > 0\}$. The contradiction with $q_3 = 0$, and the passage from $h = 0$ P_0-a.s. to identical power functions via mutual absolute continuity of Gaussian shifts, are both fine.

Suggested changes.

1.1 (Cone property used silently). The derivation of the directional inequality $d'q \geq 0$ requires $td \in \Lambda_1$ for all $t > 0$, not merely $d \in \Lambda_1$. This holds because Λ_1 is a cone (both $\mu_3 \leq \max\{\mu_1, \mu_2\}$ and $\max\{\mu_1, \mu_2\} > 0$ are preserved under positive scaling), but the proof never says so. I would add one sentence: "Since Λ_1 is a cone, $td \in \Lambda_1$ for every $t > 0$."

1.2 (Differentiation under the expectation). The right derivative of $g(t) = \mathrm{E}_0[h\,e^{td'X}]$ at $t = 0$ is computed by differentiating inside the expectation. This is justified because h is bounded ($|h| \leq 1$) and the standard Gaussian has a moment generating function finite everywhere, but a clause to that effect would help, e.g. "differentiation under the expectation is justified by dominated convergence since $|h| \leq 1$ and $\mathrm{E}_0 \exp\{t\,|d'X|\} < \infty$ for all t."

1.3 (Strictness in the final display). The display

$$\mathrm{E}_0[X_3 h] = \mathrm{E}_0[X_3 h^+] - \mathrm{E}_0[X_3 h^-] < c_3 M - c_3 M = 0$$

compresses one weak and one strict inequality into a single strict one. The strictness comes only from the second term: $X_3 > c_3$ holds a.s. on $\{h^- > 0\}$, so $\mathrm{E}_0[X_3 h^-] > c_3 M$ when $M > 0$, while $\mathrm{E}_0[X_3 h^+] \le c_3 M$ is weak. This is correct, but since the whole contradiction rests on this sign, I would spell it out in two inequalities rather than one chained display.

1.4 ("Up to null sets"). As noted in (d) above, the inclusions in the sign-support display hold everywhere, so the qualifier "up to null sets" can be dropped. Harmless either way.

1.5 (Undefined term). The statement uses "α-admissible against all level-α tests" without a definition in this note. If the term is defined in the extension plan, a pointer suffices; otherwise state it: no level-α test ψ satisfies $\mathrm{E}_\mu \psi \ge \mathrm{E}_\mu \phi_c$ for all $\mu \in \Lambda_1$ with strict inequality for some $\mu \in \Lambda_1$. Note also that the proof establishes something slightly stronger than nonexistence of a strict improvement: any weak dominator coincides with ϕ_c a.s. under every P_μ. Worth stating, since it is used verbatim in the proof of Proposition 2 ("the same continuity argument").

2 Numerical claims (Section 2 of the note)

I recomputed the calibration and the counterexample independently (adaptive quadrature for the gated acceptance probability, root-finding for b). All figures match to the displayed precision:

$$\kappa = 2.121201, \qquad b = 3.117267,$$

and at $\mu = (0.1, -5, 0.1) \in \Lambda_1$,

$$P_\mu(A_\kappa) = 0.957209, \qquad P_\mu(A_b^{\mathrm{g}}) = 0.965438,$$

so the rejection probabilities 0.042791 and 0.034562 are correct and the symmetric gate indeed fails to dominate. I also confirmed the support-function computation $h_{A_b^{\mathrm{g}}}(1,1,2) = 2b$: the bound $x_1 + x_2 + 2x_3 = (x_1 + x_3) + (x_2 + x_3) \le 2b$ is attained at $(t,\, t,\, b - t)$ for any $t \le \kappa$. The geometric obstruction argument (the common transfer cone is $\{0\}$) is correct.

One wording item: "each satisfies (2)" reads oddly since (2) is a set; "belongs to" is what is meant.

3 Proposition 2: a missing restriction on α

The Neyman–Pearson step asserts that $\phi_3 = I\{x_3 > z_\alpha\}$ *uniquely* maximizes $\mathrm{E}_0[X_3 \eta]$ over $0 \le \eta \le 1$ with $\mathrm{E}_0 \eta \le \alpha$. This is true if and only if $z_\alpha > 0$, i.e. $\alpha < 1/2$. If $\alpha \ge 1/2$, then $z_\alpha \le 0$ and the maximizer is $I\{x_3 > 0\}$, which spends mass $1/2 \le \alpha$ and does not coincide with ϕ_3: rejection mass placed where $X_3 < 0$ lowers the objective, so ϕ_3 is not optimal and the uniqueness step fails. The proposition should therefore either restrict to $\alpha \in (0, 1/2)$ (harmless in practice) or the proof should be adjusted. Also, "equality in the

preceding optimization is unique up to null sets" deserves one supporting clause: uniqueness holds because $P_0(X_3 = z_\alpha) = 0$ and any maximizer must spend the full mass α on $\{X_3 > z_\alpha\}$ when $z_\alpha > 0$.

The rest of the proof is fine: $d^{\pm} = (\pm 1, \mp 1, 1) \in \Lambda_1$, and adding the two directional inequalities gives $q_3 \geq 0$ as claimed. The follow-up remark that a conservative cutoff $c > z_\alpha$ is dominated by ϕ_3 is correct (strictly higher power at every mean, same level).

4 Proposition 4: two small gaps

4.1 (Positivity of the transfer sets). The proof asserts "Calibration gives $P_0(G_r) = P_0(D_r) > 0$." Equality is immediate, but positivity requires $b_r < c_1 + c_r$, which is not stated. It does follow from the calibration: if $b_r \geq c_1 + c_r$, then $\{x_1 \leq c_1,\ x_r \leq c_r\} \subseteq \{x_1 \leq c_1,\ x_1 + x_r \leq b_r\}$ with a containment of positive measure, so the left side of the calibration identity would strictly exceed $\Phi(c_1)\Phi(c_r)$. One sentence fixes this. (Existence and uniqueness of the finite b_r – monotone and continuous in b_r from 0 to $\Phi(c_1)$, with target strictly inside – could also be recorded, though "the unique finite b_r" is defensible as written.)

4.2 (Strictness of the transfer). The proof concludes $P_\mu(G_r) > P_\mu(D_r)$ from "the likelihood ratio is strictly larger at every point of G_r than at every point of D_r" plus equal positive null probabilities. The pointwise comparison alone gives only $\inf_{G_r} \geq \sup_{D_r}$ type control after taking limits, and the cleanest route is the explicit separating constant: with $\ell(x) := u(x_1 + x_r) - v x_r$ and $L := u b_r - v c_r$, one has $\ell > L$ on G_r (strictly, since $x_1 + x_r > b_r$ and $u > 0$) and $\ell \leq L$ on D_r. Hence

$$P_\mu(G_r) \propto \int_{G_r} e^{\ell}\, dP_0 > e^{L} P_0(G_r) = e^{L} P_0(D_r) \geq \int_{D_r} e^{\ell}\, dP_0 \propto P_\mu(D_r),$$

with the same proportionality constant on both sides. Note the inequality on D_r is weak when $v = 0$, which is why the separating constant, rather than a two-sided strict comparison, is the right tool. The remainder of the proof (factorization over the untouched coordinate s, level control by coordinatewise lowerness) is correct.

5 Minor items

5.1. "The three first Gaussian moments of h" is unusual phrasing; "the first-moment identities $\mathrm{E}_0[X_j h] = 0$, $j = 1, 2, 3$" says what is meant.

5.2. In Proposition 3, "size attained at the origin" is correct but the size is attained along the entire ray $\mu_1 = \mu_2 = \mu_3 \leq 0$ in the null; if the intended reading is uniqueness of the attainment point, that is false. As a statement of existence it is fine.

5.3. The claim that "a randomized mixture of the two single-gate dominators does dominate the rectangle" is correct: the half-half mixture is level α by linearity and its power is the average of two functions, each strictly above the rectangle's power on $\Lambda_1^{1|23}$.

5.4. The double-gate admissibility argument in Section 4 checks out internally: every finite-support direction of $A^{\mathrm{gg}}_{a,b_2,b_3}$ is a nonnegative combination of e_1, $(1, 1, 0)$, $(1, 0, 1)$, and any

such combination $(\lambda_1 + \lambda_2 + \lambda_3,\ \lambda_2,\ \lambda_3)$ with $\lambda \geq 0$ satisfies $a_2 \leq a_1$ and $a_3 \leq a_1$. I could not verify the external Proposition 1 of the extension plan itself, since that document is not included; the conclusion is conditional on that result being as described.

5.5. The closing remark of Section 3 states there is no known inadmissible exact-size test with closed, convex, coordinatewise-lower acceptance region in the two-core model. This is consistent with everything proved in the note, but as phrased it sits close to the admissibility results of Section 3; a reader may wonder whether Proposition 1's argument extends to all such regions. If it does not, one clause saying why (the tilt argument uses the rectangular faces in an essential way) would preempt the question.